Machine Learning-Based Prediction of the Specific Absorption Rate of Magnetic Nanoparticles

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Magnetic nanoparticles (MNPs) have attracted considerable interest for biomedical applications, particularly magnetic hyperthermia. Under appropriate conditions, externally applied magnetic fields can drive MNPs, while an alternating magnetic field induces heat generation can damage or destroy tumor cells. The specific absorption rate (SAR) is an important parameter for quantifying the heating performance of MNPs. In this study, we investigate the influence of the physical, chemical, and magnetic properties of MNPs, together with their surface coatings and surrounding media, on SAR prediction. Linear and nonlinear regression models are developed and compared to evaluate their predictive performance. The results demonstrate that the random forest regression model outperforms Ridge regression, achieving a coefficient of determination (R^2) of 0.98 and a root mean square error (RMSE) of 48.467 W/g. These findings highlight the potential of machine learning for predicting SAR from nanoparticle characteristics and experimental conditions, which may support the design and optimization of MNPs for magnetic hyperthermia applications.

I. INTRODUCTION:

Magnetic nanoparticles (MNPs) have a wide range of applications in modern technology, including data storage, sensors, and biomedical applications. These materials controllable sizes ranging from a few nanometres up to tens of nanometres and can exhibit physical properties that differ significantly from those of their bulk material properties[1,2]. For example, iron oxide nanoparticles with diameters of approximately 20 nm can exhibit superparamagnetic behavior at room temperature, depending on their composition, size distribution, and magnetic characteristics. This property is particularly advantageous for applications involving ferrofluids and biomedical technologies because it is biodegradable and can minimize residual magnetization after the removal of an external magnetic field[1,3].

MNPs are smaller than many biological structures, including cells, and are comparable in size to certain viruses, proteins, and genetic materials. For example, Iron oxide nanoparticles are well known for therapeutic applications, including magnetic resonance imaging (MRI) as contrast agents, targeted drug delivery, and magnetic hyperthermia for tumor treatment. In magnetic hyperthermia, coating materials of MNPs and externally applied magnetic fields can facilitate the interaction with biological targets and, under suitable conditions, guide their accumulation at specific sites, such as tumors. MNPs generate heat when exposed to an alternating magnetic field, potentially raising the temperature of tumor tissue to damage or destroy cancer cells[1,2]. The heating efficiency of MNPs can be quantified using the specific absorption rate (SAR), which represents the rate of heat generation per unit mass of magnetic material[4,5]. SAR is influenced by several factors, including the physical, chemical, and magnetic properties of nanoparticles, as well as their surface coatings[5–9]. In addition, SAR depends on the amplitude and frequency of the applied alternating magnetic field and the properties of the surrounding medium[5,10–12].

Data-driven machine learning approaches offer promising opportunities for designing application-oriented MNPs, including predicting their SAR for hyperthermia applications. Previous studies have used machine learning (ML) models to investigate and optimize metal oxide nanoparticles, including TiO_2 and iron oxide-based systems[5,13–15]. Recent studies have also explored the use of nanoparticle properties to predict SAR[5,14]. However, more research is needed to evaluate the contributions of a wider range of nanoparticle characteristics and experimental conditions to SAR prediction. Addressing these factors may help improve predictive reliability and support the development of more effective and sustainable magnetic hyperthermia technologies.

In this study, we used the physical, chemical, and magnetic properties of MNPs, together with surface coating characteristics and the properties of the surrounding medium, as input features to predict SAR. We evaluated and compared linear and nonlinear regression models to assess their predictive performance and identify the approaches best suited to capture the relationships between nanoparticle characteristics and SAR.

II. METHODOLOGY

Two machine learning regression models, Ridge regression and random forest regression, were employed to predict the SAR of MNPs. The predictive performance of the models was evaluated using three metrics: mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination (R^2). More details on the model implementation, the training procedure, and evaluation methodology are provided in the supplementary methodology file (methodology.md).

III. RESULTS AND DISCUSSIONS

FIG. 1 compares the predictive performance of the Ridge regression and random forest models. Performance was evaluated using the R^2 and RMSE. The Ridge regression model achieved an R^2 value of 0.60 and an RMSE of 195.37 W/g, indicating comparatively limited predictive performance. In contrast, the random forest model achieved an R^2 value of 0.98 and an RMSE of 48.467 W/g, demonstrating substantially better agreement between the predicted and observed SAR values. These results suggest that the random forest model is better suited to predicting SAR for the dataset considered in this study, potentially because it can capture nonlinear relationships between the input features and SAR[5].

FIG. 2 presents the Spearman correlation coefficients ($\rho = 1$) among the properties of the nanoparticle. The heat map reveals strong positive correlations (Red and orange) among the physical properties of MNPs, including the core volume, core area, and core width. Similarly, strong correlations are observed among the magnetic properties, including saturation magnetization (M_s), remanent magnetization (M_r), coercive field (H_c), and the M_r/M_s ratio. The strongest correlation in this group is observed between (M_r) and (M_r/M_s), with a (ρ = 0.97). Previous studies have also reported correlations between these parameters, consistent with the trends observed in the present study[5,16,17]. In particular, Ansari et al. (2024) reported a strong $\rho = 1$ of 0.97 involving the M_r/M_s ratio, in agreement with the correlation observed in our analysis.

FIG. 3 presents the importance feature scores for the prediction of the SAR obtained from the random forest regression model. The results indicate that H_c, the area of the nanoparticle core, and the frequency of the alternating magnetic field are among the most influential features in the prediction of the SAR.

IV. CONCLUSIONS

In conclusion, this study investigated the use of the physical, chemical, and magnetic properties of nanoparticles to predict SAR using linear and nonlinear regression models. The random forest regression model outperformed Ridge regression, achieving an R^2 of 0.98 and an RMSE of 48.467 W/g. Feature importance analysis further identified the coercive field and the frequency of the alternating magnetic field as important predictors of SAR. These findings highlight the potential of machine learning approaches for estimating SAR from nanoparticle properties and may support the design and optimization of MNPs for magnetic hyperthermia applications.

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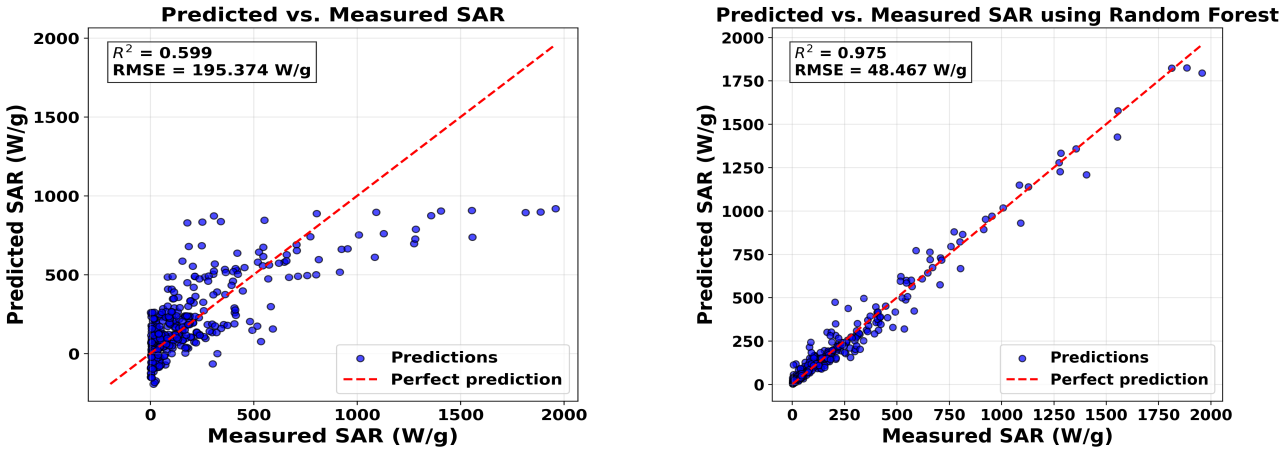
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(a) Ridge SAR

(b) RF SAR

FIG. 1: Comparison of SAR predictions obtained using Ridge regression and random forest regression. Model performance is evaluated using the coefficient of determination (R^2) and root mean square error (RMSE).

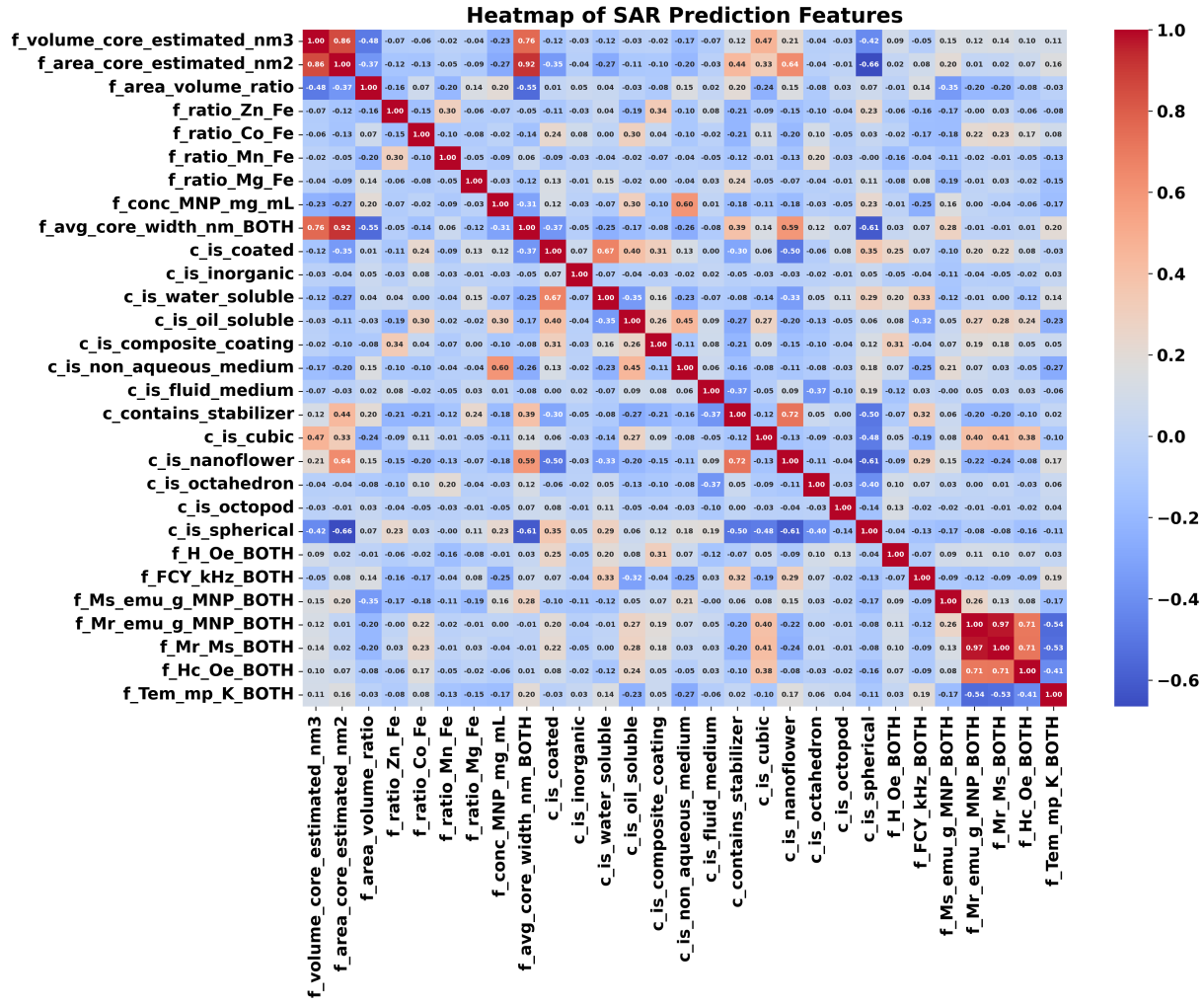


FIG. 2: Heatmap of Spearman correlation coefficients among nanoparticle features. Red indicates a perfect positive correlation ($\rho = 1$), while the color gradient transitions toward blue as the correlation decreases, with blue indicating negative correlation.

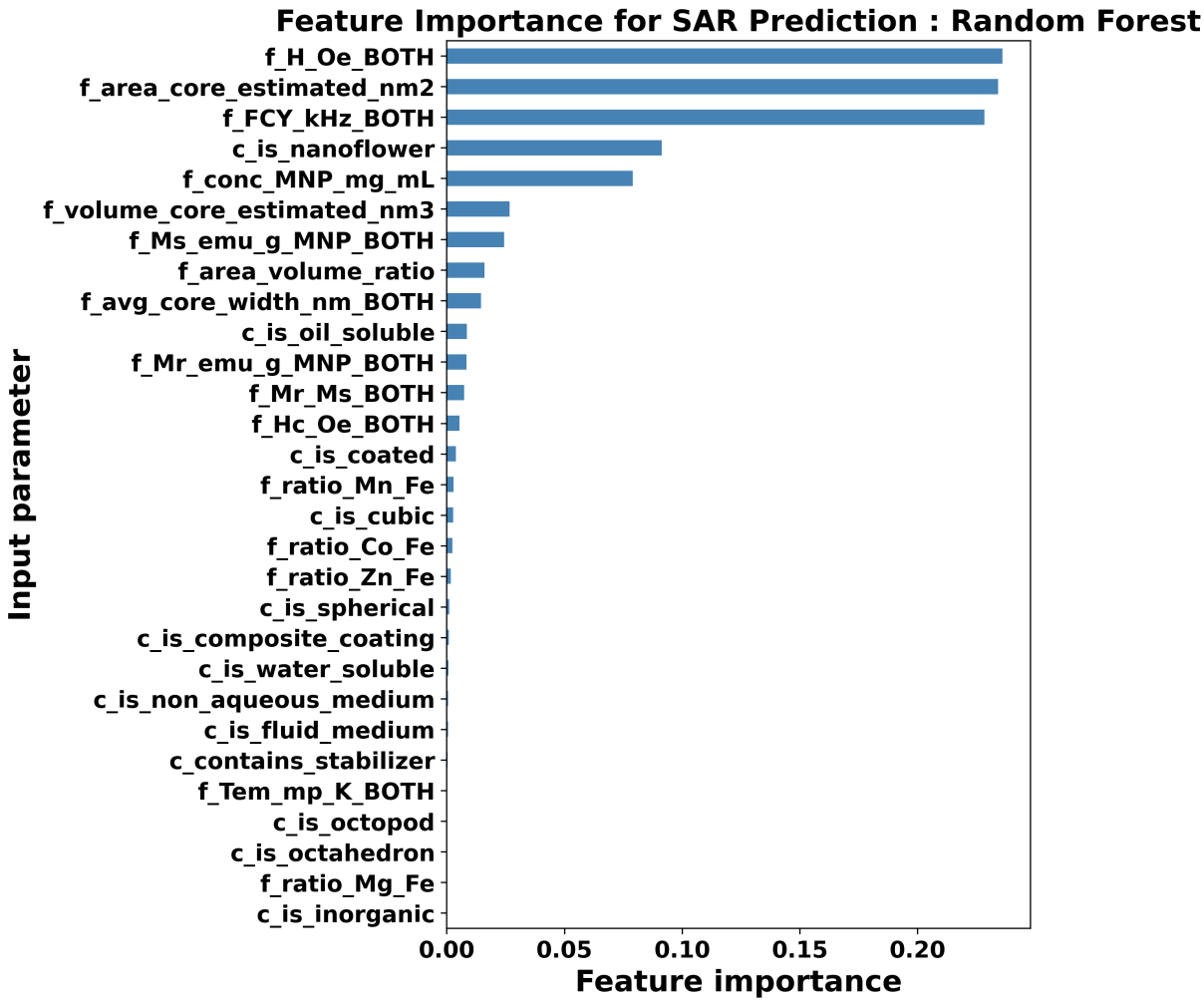


FIG. 3: Feature importance for SAR prediction derived from the random forest regression model using mean decrease in impurity (MDI). Features are ranked according to their normalized contribution to impurity reduction across the trees in the forest.